A
Project
Report On

DRIVER DROWSINESS DETECTION SYSTEM

Submitted in Partial fulfillment of the
requirements for the degree of

Bachelor of Engineering in
Computer Engineering

By

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DRIVER DROWSINESS DETECTION SYSTEM

Is a bonafide work done by

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Declaration

I declare that this written submission represents my ideas in my own words and where other's ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

Driver fatigue has emerged as one of the leading contributors to road accidents worldwide, especially in long-haul and night-time driving scenarios. In India, where road safety is a growing concern, it is estimated that drowsy driving accounts for 15–20% of all road accidents. Fatigue reduces alertness, slows reaction time, and impairs judgment, often leading to severe or fatal crashes. However, traditional methods of detecting driver fatigue—such as self-reporting or periodic monitoring—are reactive and unreliable.

To address this issue, Driver Drowsiness Detection Systems (DDDS) are being developed to identify early signs of fatigue and alert the driver in real-time. These systems primarily rely on visual indicators such as eye closure rate, blinking frequency, and yawning, which can be detected using computer vision and facial landmark recognition techniques. More advanced implementations may also incorporate physiological data (like heart rate) or vehicle-based metrics (like steering patterns), but image-based systems offer a cost-effective and non-intrusive solution suitable for mass adoption.

This report presents the design and working principles of a vision-based DDDS, which uses Eye Aspect Ratio (EAR) as the key metric for drowsiness detection. By continuously monitoring the driver’s eye movements through a webcam and analyzing facial landmarks, the system can detect prolonged eye closure—an early sign of drowsiness. Upon detection, it triggers an alert mechanism (audio or visual) to regain the driver's attention, thereby reducing the risk of accidents.

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1. Introduction

1.1 Background of the Problem (Road Safety Statistics & Importance)

Road accidents are one of the leading causes of death and injuries worldwide. According to the World Health Organization (WHO), around 1.3 million people die every year due to road traffic accidents, and millions more are seriously injured or disabled. Among these accidents, a significant percentage is caused by driver fatigue and drowsiness.

Studies reveal that drowsy driving is comparable to drunk driving in terms of impaired reaction time, reduced alertness, and poor decision-making. When drivers feel sleepy, they may experience microsleeps (brief involuntary episodes of inattention), delayed responses to road events, or even completely fall asleep while driving. This is especially common in long-distance highway driving, night shifts, or professional drivers such as truck and bus operators.

Therefore, developing a reliable **Driver Drowsiness Detection System (DDDS)** is of critical importance to enhance road safety, save lives, and prevent property damage.

1.2 Challenges in Detecting Driver Fatigue

Despite the importance of monitoring driver fatigue, detecting drowsiness in real-time is a complex task. Some major challenges include:

1. **Variability in Human Behaviour:** Different drivers show different fatigue symptoms (e.g., some blink frequently, others yawn more).
2. **Lighting Conditions:** Night driving, glare, or low light can reduce camera accuracy.
3. **Facial Occlusion:** Wearing glasses, sunglasses, masks, or head movements can make detection harder.
4. **Real-Time Processing:** Continuous video processing requires efficient algorithms to avoid lag.
5. **False Positives/Negatives:** Detecting natural blinks, head movements, or talking as drowsiness can trigger false alarms.
6. **Hardware Constraints:** System performance depends on camera quality, processing power, and robustness in real driving conditions.

1.3 Existing Solutions & Limitations

Several drowsiness detection methods are currently in use, but each has limitations:

1. **Vehicle-based Measures:** Monitoring steering wheel movements, lane deviation, or speed fluctuations.
    - Limitation: Can be inaccurate as road conditions or driving style also influence these parameters.
2. **Physiological Measures:** Using EEG, ECG, or eye-tracking sensors to monitor brain activity and fatigue levels.
    - Limitation: Intrusive, requires wearing sensors, and is uncomfortable for drivers.

3. **Camera-based Systems:** Using computer vision to monitor eye blinks, yawning, and head movements.
    - Limitation: Sensitive to lighting, face occlusion, and requires optimized real-time processing.

Although camera-based methods are non-intrusive and practical, their reliability under diverse real-world conditions remains a challenge.

1.4 Project Motivation and Objectives

The motivation for this project stems from the urgent need to reduce road accidents caused by driver fatigue. Unlike intrusive physiological methods, this project focuses on camera-based detection using computer vision and machine learning techniques to make it practical, cost-effective, and user-friendly.

Objectives:

- To detect driver drowsiness in real-time using facial landmarks (eyes, mouth, head movement).
- To calculate **Eye Aspect Ratio (EAR)** and **Mouth Aspect Ratio (MAR)** to identify prolonged eye closure and yawning.
- To trigger alert mechanisms (buzzer/alarm) when drowsiness is detected.
- To evaluate system performance with test videos/images.
- To design a solution that is scalable for future integration into vehicles.

1.5 Scope of the Project

The project is limited to software-based drowsiness detection using camera input. It does not include vehicle integration (e.g., braking or steering control) but focuses on:

- Detecting blinking patterns, yawning, and facial orientation.
- Implementing OpenCV, dlib, and facial recognition libraries for real-time detection.
- Running on a standard computer/laptop with a webcam (extendable to embedded systems like Raspberry Pi).
- Providing a low-cost, non-intrusive, and efficient solution compared to physiological sensor-based methods.

2. Literature Review

2.1 Comparative Analysis of Existing Research on Driver Drowsiness Detection System

Sr. No.	Paper Name	Year of Publication	Authors / Publications	Proposed Work	Research Gap
1	Real-Time Driver Drowsiness Detection Using Computer Vision	2023	Priya Sharma, R. Gupta, *IEEE ICMLA 2023*	Developed a real-time driver monitoring system using OpenCV and Dlib. The model detects eye aspect ratio (EAR) and triggers alarm when eyes remain closed for longer duration.	The system accuracy decreases under poor lighting and when the driver wears glasses; lacks yawning detection capability.
2	Eye and Mouth Aspect Ratio-Based Fatigue Detection	2022	Ankit Yadav, Neha Verma, *IJERT 2022*	Proposed the use of both EAR and MAR values for detecting fatigue through facial landmarks. Used thresholding technique to differentiate alert vs drowsy state.	Did not consider head movement or orientation; requires more robust multi-cue integration for higher reliability.
3	CNN-Based Drowsiness Detection Using Facial Features	2021	Rohit Singh, Sneha Menon, *IEEE ICVIP 2021*	Implemented a Convolutional Neural Network for detecting drowsy states from facial expressions. Achieved improved accuracy compared to HOG-based models.	Requires high computational resources (GPU) for real-time performance; not suitable for low-end systems or embedded hardware.

2.2 Existing System

Traditional driver monitoring approaches rely mainly on vehicle-based or physiological sensors such as steering wheel motion, ECG/EEG readings, or heart rate sensors. While these methods can indicate fatigue, they have several limitations:

- Intrusive: Require physical contact sensors which are uncomfortable for drivers.
- Expensive: High hardware cost and complex setup.
- Inflexible: Difficult to adapt for general-purpose vehicles.
- Limited accuracy: Physiological variations differ among individuals.

2.3 Limitations of Existing System

- Unable to detect fatigue without direct physical contact.
- Slow response due to sensor lag or calibration errors.
- Fails to operate accurately in real-world driving conditions such as low light or partial face occlusion.
- High installation and maintenance cost for sensors or specialized equipment.
- No mechanism for real-time visual confirmation or automated alerting.

2.4 Proposed System

The proposed system uses camera-based computer vision techniques to non-intrusively monitor the driver's facial features.

- Uses OpenCV and Dlib libraries for face and landmark detection.
- Computes Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) to detect prolonged eye closure and yawning.
- Integrates CNN/HOG models for robust face detection under different lighting conditions.
- Implements an alert mechanism that activates an alarm, message, or vibration when drowsiness is detected.

2.5 Benefits of Proposed System

- Non-Intrusive: Requires only a camera; no physical sensors.
- Real-Time Processing: Detects drowsiness instantly through facial landmarks.
- Cost-Effective: Can be implemented on low-cost hardware such as Raspberry Pi or laptop webcam.
- Scalable: Can be integrated into cars, trucks, or fleet systems.
- Accurate & Efficient: Uses multiple indicators (eyes, mouth, head) for reliable detection.
- Improved Safety: Reduces road accidents caused by fatigue by providing early alerts.

3. Software Requirements

3.1 Operating System

Supported: Windows 10/11 (64-bit), Linux (Ubuntu 20.04+), macOS

Reason: The required libraries (OpenCV, dlib, face_recognition) are cross-platform, but Windows requires additional setup (Visual C++ Build Tools, CMake). Linux is often preferred for easier compilation of dlib.

3.2 Programming Language & Dependencies

- **Python Version:** 3.8 – 3.10 (recommended for compatibility)
  - Reason: Some libraries like dlib and face_recognition may not yet support the newest Python versions.
- **Libraries Required:**
  - **OpenCV (cv2):** For image processing, video capture, and drawing bounding boxes.
  - **dlib:** For facial landmark detection (eyes, mouth, etc.).
  - **face_recognition:** Wrapper around dlib for face detection and recognition.
  - **NumPy:** For array operations and numerical computations.
  - **SciPy:** For scientific computations (e.g., distance calculations).
  - **Matplotlib/Seaborn:** For visualizing EAR/MAR graphs and results.
  - **Pygame:** For Alarm

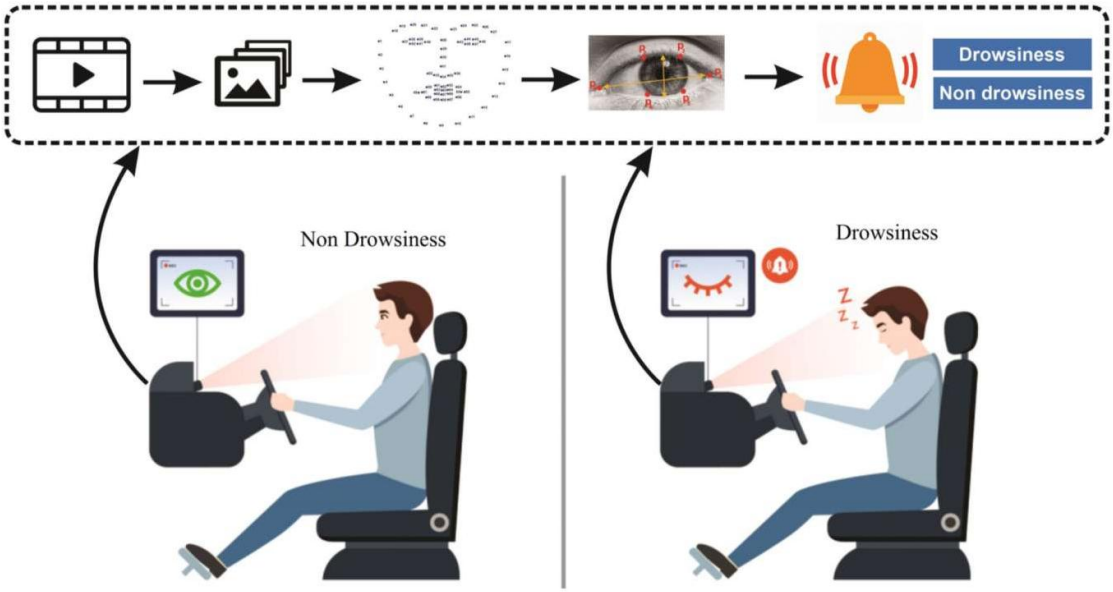
3.3 Integrated Development Environment (IDE) / Notebook

- **Jupyter Notebook (preferred):**
  - Provides an interactive environment for step-by-step testing, debugging, and visualization of results.
- **Alternative IDEs:** PyCharm, VS Code, or Spyder
  - Better suited for writing production-ready, modular code.

4. System Design

The **Driver Drowsiness Detection System (DDDS)** is designed as a modular pipeline that captures a driver's facial video feed, processes facial landmarks in real time, applies detection algorithms, and triggers alerts if drowsiness symptoms are observed.

4.1 Architecture Diagram



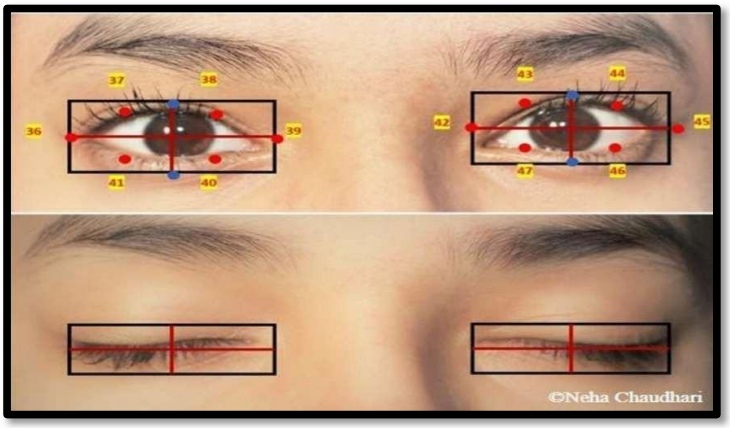
Explanation:

- **Input:** A webcam continuously streams the driver's face in real time.
- **Processing:** The system extracts frames, detects facial landmarks, computes metrics (EAR, MAR), and applies CNN/HOG face detection for robustness.
- **Output:** If drowsiness is detected, the system triggers an alarm**.**

4.2 Algorithms / Models Used

(i) Eye Aspect Ratio (EAR) – Blink Detection

- EAR is a measure of the ratio between distances of vertical and horizontal eye landmarks.
- Formula: $EAR = \frac{(P2-P6)+(P3-P5)}{2*(P1-P4)}$



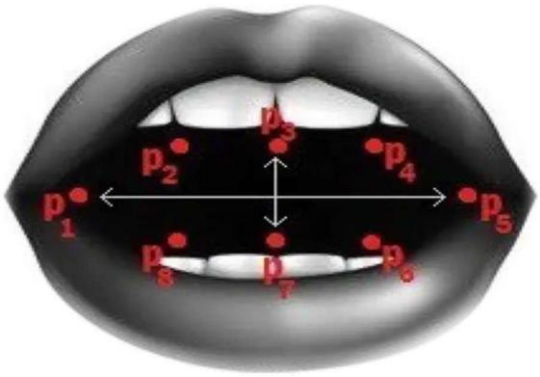
Logic:

- If EAR < threshold for several consecutive frames → eyes are closed → possible drowsiness.

(ii) Mouth Aspect Ratio (MAR) – Yawning Detection

- MAR uses distances between upper and lower lip landmarks.

  Formula: $MAR = \frac{(P3-P9)+(P4-P8)+(P5-P7)}{2*(P1-P6)}$



Logic:

- If MAR > threshold for several consecutive frames → yawning detected → sign of drowsiness.

(iii) Face Detection (HOG vs CNN)

- **HOG (Histogram of Oriented Gradients):**
  - Lightweight, CPU-friendly, works well in good lighting.
  - May fail in low-light or with occlusion.
- **CNN (Convolutional Neural Networks):**
  - More accurate, robust to pose variations.
  - Requires GPU/strong CPU for real-time performance.
- **Choice:** HOG can be used for real-time on low-end systems; CNN for higher accuracy.

(iv) Alert System Logic

- If EAR < threshold for T frames **OR** MAR > threshold → system activates alarm.
- Possible outputs:
  - **Audio Alert:** Beep sound / buzzer to wake driver.
  - **Visual Alert:** Warning message on screen.
  - **Logging:** Record event (time, duration of drowsiness).

5. Methodology

5.1 Step-by-step explanation of implementation

Step 1: Capturing Video Stream using Camera

- A webcam or in-car camera continuously captures video frames of the driver.
- The frames are processed in real-time using **OpenCV**.
- Each frame is passed through the pipeline for analysis without noticeable delay.
- **Advantage**: Provides continuous monitoring without physical sensors.

Step 2: Detecting Face and Landmarks (Eyes, Mouth)

- **Face Detection**:
  - HOG (Histogram of Oriented Gradients) for lightweight processing.
  - CNN (Convolutional Neural Network) for higher accuracy in varied conditions.
- **Landmark Extraction**:
  - Dlib's pre-trained 68-point facial landmark detector is used.
  - Key landmarks:
    - Eyes (6 points each).
    - Mouth (20 points).
    - Nose & jawline (optional, for head pose).

Step 3: Calculating EAR and MAR Thresholds

- Eye Aspect Ratio (EAR) is computed from eye landmarks:
- $\text{EAR} = \frac{(P2-P6)+(P3-P5)}{2*(P1-P4)}$

- EAR remains constant when eyes are open.
- EAR decreases when eyes close.
- If EAR < 0.25 for ~48 consecutive frames → drowsiness alert.
- Mouth Aspect Ratio (MAR) is computed from mouth landmarks:

  $\text{MAR} = \frac{(P3-P9)+(P4-P8)+(P5-P7)}{2*(P1-P6)}$

- MAR increases during yawning.
- If MAR > 0.6–0.7 for several frames → yawning alert.

Step 4: Identifying Drowsiness Conditions

- **Long Blinks**: EAR drops below threshold continuously → driver's eyes are closed.
- **Yawns**: MAR increases beyond threshold → driver is fatigued.

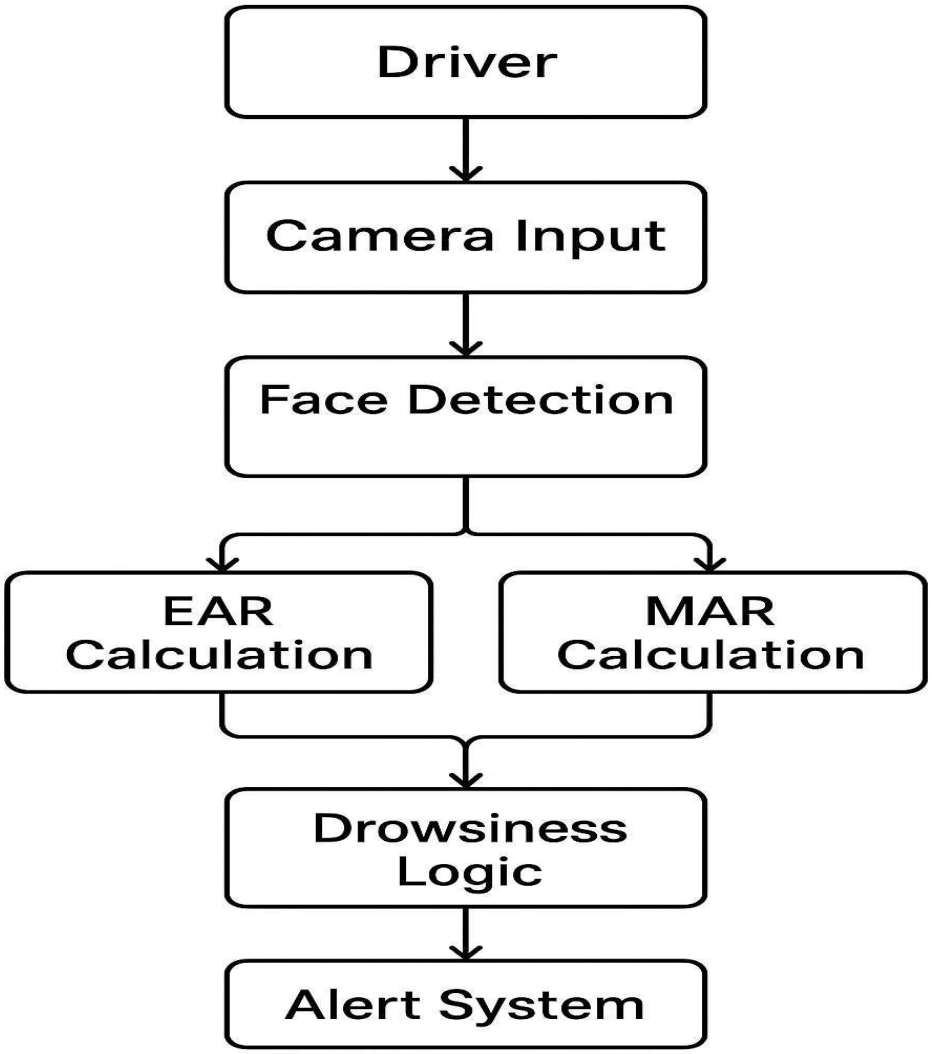
Step 5: Triggering Alerts

If drowsiness is detected, the system initiates alerts:

- **Audio Alert**: Buzzer or voice message.
- **Visual Alert**: On-screen popup or warning indicator.

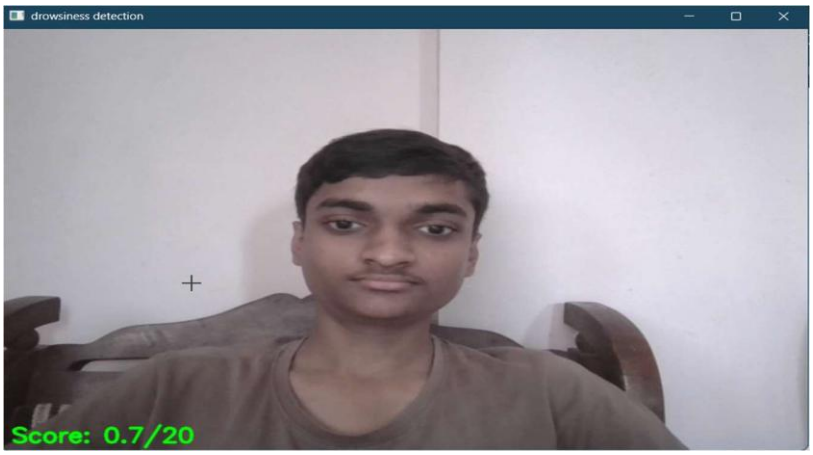
The alert ensures the driver is notified immediately, reducing accident risks.

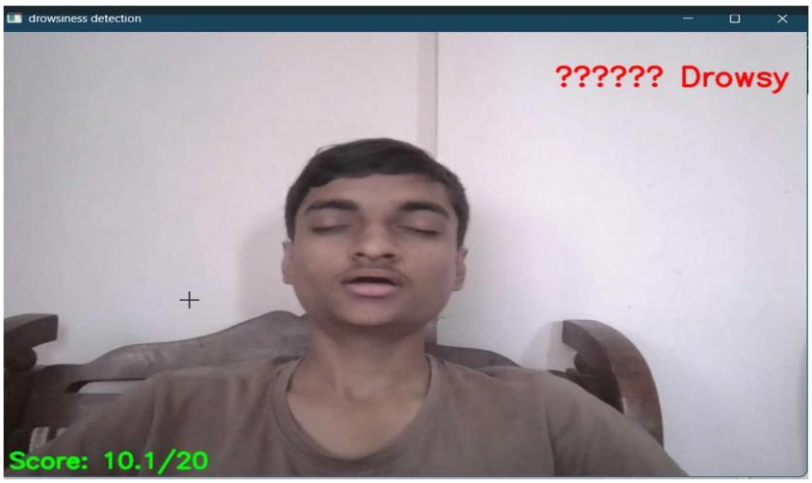
5.2 Flowchart of the Process



6. Result and Analysis

6.1 Sample Output





6.2 Discussion of Effectiveness and Limitations

- **Effectiveness**
  - The system is effective in detecting prolonged eye closures, frequent yawning, and head tilts, which are strong indicators of drowsiness.
  - Real-time alerts can potentially prevent accidents.
  - Works well in controlled environments and on standard webcams.
- **Limitations**
  - Performance decreases in low-light environments without IR cameras.
  - Sunglasses or partially covered faces reduce accuracy.
  - Micro-sleeps shorter than 1 second may be missed.
  - The system depends heavily on camera quality and processing power.

7. Conclusion

7.1 Summary of Achievements

The Driver Drowsiness Detection System successfully demonstrates how computer vision and machine learning techniques can be applied to enhance road safety. Key achievements include:

- Real-time detection of driver fatigue indicators such as prolonged eye closure (blinking rate), yawning, and head tilting using facial landmark detection.
- Implementation of Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) algorithms, which provide reliable thresholds for identifying drowsiness and yawning episodes.
- Integration of CNN/HOG-based face detection models for robustness under varying conditions.
- Development of an alert mechanism (alarm sound, vibration, or on-screen warning) that notifies the driver immediately when drowsiness is detected.
- A modular and scalable architecture that can be extended with additional sensors (heart rate, steering patterns, etc.) in future work.

7.2 Impact on Road Safety

Drowsiness is one of the leading causes of road accidents worldwide, often leading to severe injuries and fatalities. The proposed system directly addresses this challenge by:

- Reducing accident risks by providing early warnings before the driver completely loses alertness.
- Supporting long-distance drivers and night-shift workers, who are most prone to fatigue-related accidents.
- Offering a low-cost and non-intrusive solution that only requires a camera and a computer, making it practical for wide adoption.
- Demonstrating the potential for integration into advanced driver assistance systems (ADAS), contributing toward safer autonomous and semi-autonomous vehicles.

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